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Bio Medical Signal Processing

BOE083

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** ALL | **Semester:** 8 | **Credits:** 3 | **L-T-P:** 3-0-0

Scheme: AKTU B.Tech 4th year, Open Elective-III, 2025-26 onwards

UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5
Introduction to Biomedical Signals	ECG Signal Processing	ECG Data Reduction Techniques	EEG Signal Processing and Spectral Estimation	Evoked Potentials, Adaptive Filtering and Wavelets

Original revision notes written for the AKTU syllabus. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: the theory, algorithms step by step, and tables.
3. **Solved examples**: heart rate, QTc, notch filter, Pan-Tompkins filters, AZTEC, turning point, Huffman coding, frequency resolution, AR model, signal averaging, LMS update.
4. **Key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (3 hours):** Unit 1 (signal types, ECG/EEG/EMG/ERG basics, acquisition problems) and Unit 2 (ECG measurements, QRS detection, noise removal, arrhythmia).
- **Afternoon (3 hours):** Unit 3 (turning point, AZTEC, Fan, Huffman, run-length) and Unit 4 (EEG, sleep, epilepsy, spectral estimation).
- **Evening (2 hours):** Unit 5 (signal averaging, LMS, noise cancelling, wavelets), then the "Last-hour checklist".
- **In the exam:** draw the ECG waveform with labels, the Pan-Tompkins block diagram, and the adaptive noise canceller. Write difference equations neatly.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

UNIT 1 Introduction to Biomedical Signals

✍️ 2-mark definitions

- **Biomedical signal:** a signal produced by a physiological process (electrical, mechanical, chemical, acoustic) that carries information about the state of the body, e.g. ECG, EEG, EMG, blood pressure, heart sounds.
- **Bioelectric potential:** a voltage produced by the movement of ions (Na^+, K^+, Ca^{2+}) across cell membranes; the basis of ECG, EEG and EMG.
- **Action potential:** the rapid depolarization and repolarization of an excitable cell membrane (resting about −90 mV in heart muscle, peak about +20 mV).
- **ECG (electrocardiogram):** the record of the heart's electrical activity from body-surface electrodes.
- **EEG (electroencephalogram):** the record of the brain's electrical activity from scalp electrodes.
- **EMG (electromyogram):** the record of the electrical activity of skeletal muscles.
- **ERG (electroretinogram):** the record of the electrical response of the retina to a light stimulus.
- **Artifact:** an unwanted signal not from the organ of interest, e.g. motion, power line interference, muscle noise, electrode contact noise.
- **Deterministic vs random signal:** a deterministic signal can be described exactly by a formula; a random (stochastic) signal can only be described statistically (most biosignals are random/non-stationary).
- **Stationary signal:** a signal whose statistical properties (mean, variance, autocorrelation) do not change with time.

💡 Core concepts

Classification of biomedical signals

Basis	Classes
Origin	Bioelectric (ECG, EEG, EMG, EOG, ERG), biomechanical (respiration, motion), bioacoustic (heart sounds, PCG), biochemical (blood gases, glucose), biomagnetic (MEG), bio-optical (SpO_2, PPG), bioimpedance
Nature	Continuous vs discrete; deterministic vs random; periodic (ECG roughly), quasi-periodic, transient (evoked potentials); stationary vs non-stationary
Dimension	1-D signals (ECG), 2-D images (X-ray, MRI), 3-D volumes

Characteristics of common biosignals

Signal	Amplitude	Frequency range	Electrodes / sensor
ECG	0.5–4 mV	0.05–100 Hz (diagnostic), 0.5–40 Hz (monitor)	Ag/AgCl surface electrodes
EEG	10–100 µV (scalp)	0.5–100 Hz	10–20 system scalp electrodes

Signal	Amplitude	Frequency range	Electrodes / sensor
EMG	0.1–5 mV	20–500 Hz (surface)	Surface or needle electrodes
ERG	0.5 µV–1 mV	0.2–200 Hz	Corneal (contact lens) electrode
EOG	50–3500 µV	DC–50 Hz	Electrodes around the eyes

ECG basics: generated by the heart's conduction system: SA node (pacemaker, 60–100 beats/min) → atria → AV node (delay) → bundle of His → bundle branches → Purkinje fibres → ventricles. Waves: **P** (atrial depolarization), **QRS** (ventricular depolarization), **T** (ventricular repolarization), sometimes **U**. Standard 12-lead ECG: 3 bipolar limb leads (I, II, III — Einthoven's triangle), 3 augmented leads (aVR, aVL, aVF), 6 chest leads (V1–V6).

EEG basics: summed post-synaptic potentials of cortical neurons; electrodes in the international 10–20 system. Rhythms: **delta** 0.5–4 Hz (deep sleep), **theta** 4–8 Hz (drowsiness), **alpha** 8–13 Hz (relaxed, eyes closed), **beta** 13–30 Hz (alert, thinking), **gamma** > 30 Hz.

EMG basics: motor unit action potentials (MUAPs) summed; amplitude grows with force; used in nerve-muscle disease diagnosis, prosthesis control and rehabilitation.

ERG basics: a-wave (negative, photoreceptors), b-wave (positive, bipolar/Müller cells); used to diagnose retinal diseases.

Difficulties in acquiring biosignals

- Very small amplitudes (µV–mV) → need high-gain, high-CMRR instrumentation amplifiers.
- Noise and interference: 50/60 Hz power line, motion artifacts, baseline wander (respiration), muscle noise (EMG in ECG), electrode polarization and contact impedance.
- Inaccessibility of the source (signals measured indirectly through tissue).
- Variability between and within subjects; non-stationarity.
- Physiological interactions (one system affects another).
- Patient safety (leakage currents, isolation).
- Effect of the instrument on the system and patient discomfort.

Computers in biomedical signal analysis: acquisition and digitization (ADC), filtering and noise removal, feature extraction (heart rate, intervals), pattern classification (arrhythmia, seizure), data compression and storage, telemedicine, real-time monitoring and alarms, decision support, and **image reconstruction** (CT from projections by filtered back-projection; MRI by inverse Fourier transform).

🧮 Solved examples

Example 1: Sampling rate. An ECG needs a bandwidth of 0.05–100 Hz. Nyquist rate = 2 × 100 = 200 Hz; in practice 250–500 Hz (MIT-BIH database uses 360 Hz).

Example 2: Amplifier gain. An EEG of 50 µV must be raised to 1 V for an ADC. Gain = 1 / (50 × 10^{-6}) = **20,000** (86 dB).

Example 3: ADC resolution. A 12-bit ADC with ±5 mV input range (10 mV span) has resolution = 10 mV / 4096 ≈ **2.44 µV**, adequate for ECG.

Key points to remember

- ECG: mV, 0.05–100 Hz; EEG: µV, 0.5–100 Hz; EMG: 20–500 Hz.
- P = atrial depolarization, QRS = ventricular depolarization, T = ventricular repolarization.
- EEG bands: delta, theta, alpha, beta, gamma.
- Acquisition difficulties: small signals, noise, inaccessibility, variability, safety.

⭐ Important questions

1. Classify biomedical signals with examples.
2. Explain the origin of the ECG and its waves. Explain the 12-lead system.
3. Explain EEG, EMG and ERG signals and their characteristics.
4. What are the difficulties in acquiring biomedical signals?
5. Explain the role of computers in biomedical signal analysis and image reconstruction.

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UNIT 2 ECG Signal Processing

✍️ 2-mark definitions

- **RR interval:** time between successive R peaks; heart rate (beats/min) = 60 / RR (s).
- **PR interval:** start of P to start of QRS (0.12–0.20 s); AV conduction time.
- **QRS duration:** width of the QRS complex (0.06–0.10 s; > 0.12 s is abnormal).
- **QT interval:** start of QRS to end of T (about 0.35–0.44 s), corrected for heart rate as QTc = QT / √RR (Bazett).
- **ST segment:** end of S to start of T; elevation or depression indicates myocardial ischemia or infarction.
- **QRS detection:** automatic location of each QRS complex, the first step in almost all ECG analysis.
- **Baseline wander:** slow drift of the ECG baseline (< 0.5 Hz) due to respiration and electrode movement.
- **Power line interference:** 50 Hz (India) or 60 Hz (USA) sinusoidal noise from mains wiring.
- **Arrhythmia:** any abnormal heart rhythm (too fast, too slow, irregular, or abnormal beats).
- **Holter monitor:** a portable ECG recorder worn for 24–48 hours to catch intermittent arrhythmias.

💡 Core concepts

Amplitude and interval measurement: detect R peaks, then locate onsets and offsets of P, QRS, T by slope thresholds or templates; measure amplitudes from the isoelectric baseline (PR segment). Derived values: heart rate, RR variability, PR, QRS, QT/QTc, ST level (measured at J point + 60–80 ms).

QRS detection approaches: amplitude and first-derivative thresholds; first and second derivative combined; template matching; filter banks; wavelet-based; and the most famous, **Pan–Tompkins**.

Pan–Tompkins algorithm (1985, fs = 200 Hz) — draw the block diagram:

1. **Band-pass filter (5–15 Hz)** = integer low-pass + high-pass, to keep QRS energy and reject P/T waves, baseline and muscle noise. - Low-pass: $y(n) = 2y(n-1) - y(n-2) + x(n) - 2x(n-6) + x(n-12)$ (cut-off ≈ 11 Hz, delay 5 samples). - High-pass: $y(n) = 32x(n-16) - [y(n-1) + x(n) - x(n-32)]$ (cut-off ≈ 5 Hz, delay 16 samples).
2. **Derivative:** $y(n) = (1/8)[2x(n) + x(n-1) - x(n-3) - 2x(n-4)]$, to highlight steep QRS slopes.
3. **Squaring:** $y(n) = x(n)^2$, makes all values positive and emphasizes large slopes.
4. **Moving-window integration:** $y(n) = (1/N)[x(n-N+1) + \ldots + x(n)]$, with N = 30 samples (150 ms), gives a waveform feature for QRS width.
5. **Adaptive thresholds and decision rules:** two thresholds on both the filtered and integrated signals updated with running estimates of signal and noise peaks; search-back if no QRS in 166% of the average RR; 200 ms refractory period; T-wave discrimination by slope.

ST-segment analysis: measure ST deviation relative to the PR baseline at J + 60/80 ms; elevation > 1 mm (0.1 mV) in two contiguous leads suggests infarction; depression suggests ischemia; trend ST over time in stress tests and ICU monitors.

Removing baseline wander

- High-pass filter with cut-off about 0.5 Hz (linear phase FIR or forward-backward IIR to avoid distorting the ST segment).
- Median filters (200 ms then 600 ms) to estimate and subtract the baseline.
- Cubic spline through PR-segment points, subtracted.
- Wavelet decomposition, removing the lowest approximation band.

Removing power line interference

- **Notch filter** with zeros on the unit circle at $\pm\omega_o = \pm 2\pi \cdot f_o / f_s$ and poles near them (radius $r < 1$) for a narrow notch: $H(z) = (1 - 2\cos\omega_o z^{-1} + z^{-2}) / (1 - 2r\cos\omega_o z^{-1} + r^2 z^{-2})$.
- Adaptive noise cancelling with a reference from the mains (Unit 5).
- Good hardware: high CMRR amplifier, driven right leg circuit, shielding, short leads.

Arrhythmia analysis: extract features (RR, QRS width, morphology, P presence) → classify beats (normal N, premature ventricular contraction PVC, atrial premature APC, bundle branch block) → classify rhythm.

Rhythm	Key features
Normal sinus	Rate 60–100, regular, P before each QRS, PR normal
Sinus bradycardia / tachycardia	Rate < 60 / > 100, otherwise normal
Atrial fibrillation	Irregularly irregular RR, no clear P waves, fibrillatory baseline
Atrial flutter	Saw-tooth flutter waves (~300/min), regular ventricular response
PVC	Early wide bizarre QRS (> 0.12 s), no P, compensatory pause
Ventricular tachycardia	Run of ≥ 3 PVCs, rate > 100, wide QRS
Ventricular fibrillation	Chaotic waveform, no QRS, fatal without defibrillation
Heart block	Prolonged PR (1st degree), dropped beats (2nd), AV dissociation (3rd)

Portable arrhythmia monitors: Holter recorders, event recorders, implantable loop recorders, wearable patches and smartwatches. Needs: low power, real-time QRS detection, data compression (Unit 3), memory, alarms, wireless transmission to doctors.

🧮 Solved examples

Example 1: Heart rate. ECG sampled at 360 Hz; two R peaks are 288 samples apart. RR = 288/360 = 0.8 s. HR = 60/0.8 = **75 beats/min**.

Example 2: QTc. QT = 0.36 s, RR = 0.81 s. QTc = 0.36 / √0.81 = 0.36 / 0.9 = **0.40 s** (normal, < 0.44 s).

Example 3: Simple 50 Hz notch. $f_s = 200$ Hz. $\omega_o = 2\pi \times 50/200 = \pi/2$. Zeros at $e^{\pm j\pi/2} = \pm j$. $H(z) = 1 + z^{-2}$. Normalize DC gain to 1: **$y(n) = [x(n) + x(n-2)] / 2$**. Check: at 50 Hz, $z^{-2} = e^{-j\pi} = -1$, so $H = 0$. ✓ (A pole pair at $r = 0.95$ would narrow the notch.)

Example 4: Integration window. At $f_s = 200$ Hz, a 150 ms window has $N = 0.150 \times 200 =$ **30 samples**. At 360 Hz it would be 54 samples.

Example 5: Derivative output. x(n) = 10, x(n−1) = 8, x(n−3) = 3, x(n−4) = 1. y(n) = (1/8)(20 + 8 − 3 − 2) = 23/8 = **2.875**.

Key points to remember

- HR = 60/RR; PR 0.12–0.20 s; QRS < 0.12 s; QTc = QT/√RR.
- Pan–Tompkins: band-pass (5–15 Hz) → derivative → squaring → moving-window integration (150 ms) → adaptive thresholds.
- Baseline wander: HPF 0.5 Hz, median filter, spline. Power line: notch at 50 Hz.
- AF = irregular, no P; PVC = early wide QRS, compensatory pause.

⭐ Important questions

1. Explain amplitude and interval measurement in ECG and their clinical meaning.
2. Explain the Pan–Tompkins QRS detection algorithm with block diagram and difference equations.
3. Explain methods for removing baseline wander and power line interference. Design a 50 Hz notch filter.
4. Explain ST-segment analysis and arrhythmia analysis.
5. Write a note on portable arrhythmia monitors.

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UNIT 3 ECG Data Reduction Techniques

✍️ 2-mark definitions

- **Data reduction (compression):** representing a signal with fewer bits while keeping the clinically important information.
- **Compression ratio (CR):** original data size / compressed data size.
- **PRD (percent root-mean-square difference):** PRD = 100 × √(Σ(x − x̃)² / Σx²); measures reconstruction error.
- **Lossless compression:** exact reconstruction possible (Huffman, run-length).
- **Lossy compression:** approximate reconstruction with higher CR (turning point, AZTEC, Fan).
- **Turning point (TP) algorithm:** keeps one of every two samples, choosing the one that preserves turning points (peaks and troughs); CR fixed at 2:1.
- **AZTEC (Amplitude Zone Time Epoch Coding):** converts the ECG into horizontal lines (plateaus) and slopes, storing (length, amplitude) pairs.
- **Fan algorithm:** a straight-line approximation that keeps extending a "fan" of permitted slopes within a tolerance ε, storing only the end points of each line.
- **Huffman coding:** a variable-length prefix code that gives shorter codewords to more probable symbols, approaching the entropy.
- **Run-length encoding (RLE):** replaces a run of identical values by (value, count).

💡 Core concepts

Why compress ECG: a 24-hour, 3-lead Holter at 250 Hz and 12 bits gives about 3 × 250 × 12 × 86,400 / 8 ≈ 97 MB; compression saves memory, transmission time and cost (telemedicine).

Turning point algorithm

1. Take the saved sample x_o and the next two samples x_1, x_2.
2. Compute $s_1 = \text{sign}(x_1 - x_o)$, $s_2 = \text{sign}(x_2 - x_1)$.
3. If $s_1 \times s_2 < 0$ (x_1 is a turning point), save x_1; otherwise save x_2.
4. The saved sample becomes the new x_o; repeat. - CR = 2:1 always; preserves peaks; distorts timing slightly (non-uniform effective sampling).

AZTEC algorithm

- **Plateau stage:** keep adding samples while (max − min) of the current group ≤ threshold (aperture) and length < limit; when broken, save the plateau as (length, average value).
- **Slope stage:** consecutive short plateaus (fewer than 3 samples) are merged into a slope, saved as (−length, final value).
- Reconstruction gives a step-like signal; smoothing (parabolic filter) improves appearance.
- CR about 10:1, but it distorts P and T waves (step-like), and the reconstruction is discontinuous.
- **CORTES** (Coordinate Reduction Time Encoding System) = AZTEC for isoelectric parts + TP for QRS: better fidelity.

Fan algorithm

1. Start at a saved point; draw upper and lower slopes to the next sample ± ε (a fan).
2. For each new sample, narrow the fan to the tighter of the old limits and the new ones.
3. When a new sample falls outside the fan, save the previous sample as the line end and start a new fan from it.
4. Reconstruct by straight lines between saved points. CR depends on ε and the signal; typical 3–10.

Huffman coding

1. Take the first difference of the ECG (most differences are small, near zero → skewed distribution).
2. List symbols with probabilities; repeatedly merge the two least probable into a node until one remains.
3. Assign 0/1 on branches; read the code from root to leaf.
4. **Modified Huffman:** frequent symbols get Huffman codes; rare symbols are sent with a short "escape" prefix + raw value, keeping the code table small. - Average length $\bar{L} = \Sigma\, p_i\, l_i$; entropy $H = -\Sigma\, p_i \log_2 p_i$; $H \le \bar{L} < H + 1$. Lossless.

Run-length coding: good for flat isoelectric segments of the difference signal (long runs of 0); lossless.

Comparison

Method	Type	Typical CR	Remarks
Turning point	Lossy	2	Keeps peaks, simple
AZTEC	Lossy	~10	Step distortion of P/T
CORTES	Lossy	~5	Better QRS fidelity
Fan	Lossy	3–10	Piecewise linear, good fidelity
Huffman (on differences)	Lossless	2–3	Needs code table
Run-length	Lossless	Data dependent	Good for long flat runs

🧮 Solved examples

Example 1: Turning point. Samples: $x_0 = 10$, $x_1 = 14$, $x_2 = 12$. $s_1 = \text{sign}(4) = +$, $s_2 = \text{sign}(-2) = -$; product < 0, so **save $x_1 = 14$** (a peak). Next: $x_0 = 14$, $x_1 = 11$, $x_2 = 9$: $s_1 = -$, $s_2 = -$; product > 0, so **save $x_2 = 9$**.

Example 2: AZTEC plateaus (threshold 4). Samples: 10, 11, 10, 12, 30, 31, 30, 29, 10.

- 10, 11, 10, 12: range 2 ≤ 4. Adding 30 makes range 20, so save plateau (**4, 10.75**).
- 30, 31, 30, 29: range 2; adding 10 breaks it → save (**4, 30**).
- 10 alone at the end → (1, 10).
- 9 samples become 3 pairs = 6 numbers: CR = 9/6 = **1.5** (real ECG with long flat regions gives about 10).

Example 3: Huffman code for difference symbols. P(0) = 0.4, P(+1) = 0.2, P(−1) = 0.2, P(+2) = 0.1, P(−2) = 0.1.

- Merge +2, −2 → 0.2 (node X). Merge +1, −1 → 0.4 (node Y). Merge X and Y → 0.6. Merge with 0 → 1.0.

- Codes: 0 → `0`; +1 → `100`; −1 → `101`; +2 → `110`; −2 → `111`.
- $\bar{L}$ = 0.4 × 1 + 0.6 × 3 = **2.2 bits/symbol**.
- Entropy H = 0.4(1.322) + 2 × 0.2(2.322) + 2 × 0.1(3.322) ≈ **2.12 bits**. Efficiency = 2.12/2.2 = 96%.
- Versus a 3-bit fixed code: CR = 3/2.2 ≈ **1.36**.

Example 4: Run-length. Difference sequence 0 0 0 0 0 0 3 −2 0 0 0 0 → (0,6) (3,1) (−2,1) (0,4).

Example 5: PRD. Original [2, 4, 6], reconstructed [2, 5, 6]. PRD = 100 × √(1/(4 + 16 + 36)) = 100 × √(1/56) ≈ **13.4%**.

Key points to remember

- CR = original/compressed; PRD measures error.
- TP: fixed 2:1; keep x_1 if $(x_1 - x_0)(x_2 - x_1) < 0$, else x_2.
- AZTEC: plateaus (length, value) and slopes; CR ~10; step distortion. CORTES = AZTEC + TP.
- Fan: tolerance ε, straight lines.
- Huffman on first differences; $H \le \bar{L} < H + 1$; modified Huffman uses escape codes.

⭐ Important questions

1. Why is ECG data reduction needed? Define CR and PRD.
2. Explain the turning point algorithm with an example.
3. Explain the AZTEC algorithm and its drawbacks. What is CORTES?
4. Explain the Fan algorithm.
5. Explain Huffman, modified Huffman and run-length coding with an example.

UNIT 4 EEG Signal Processing and Spectral Estimation

✍️ 2-mark definitions

- **EEG rhythm:** a band of EEG frequencies associated with a brain state (delta, theta, alpha, beta, gamma).
- **Sleep stages:** wake, N1, N2, N3 (non-REM) and REM, scored from EEG, EOG and EMG (polysomnography).
- **Sleep spindle:** a burst of 12–14 Hz (11–16 Hz) activity lasting 0.5–2 s, typical of stage N2.
- **K-complex:** a large biphasic wave in stage N2.
- **Epilepsy:** a brain disorder with recurrent seizures caused by abnormal synchronous neuron firing; EEG shows spikes, sharp waves and spike-and-wave complexes.
- **Power spectral density (PSD):** the distribution of signal power over frequency; the Fourier transform of the autocorrelation function.
- **Periodogram:** PSD estimate $P(f) = (1/N)|X(f)|^2$ from the DFT of N samples.
- **Parametric spectral estimation:** fitting a model (AR, MA, ARMA) to the data and computing the PSD from the model parameters.
- **AR (autoregressive) model:** $x(n) = -\Sigma\, a_k\, x(n-k) + e(n)$; an all-pole model.
- **Maximum entropy method (MEM, Burg):** the spectral estimate that is most random (maximum entropy) while matching the known autocorrelation values; equivalent to AR estimation.

💡 Core concepts

EEG analysis tasks: artifact removal (eye blinks, muscle, line noise), rhythm and band power analysis, event detection (spikes, spindles), sleep staging, seizure detection, brain–computer interfaces, depth of anaesthesia monitoring.

Sleep EEG

Stage	EEG features
Wake (eyes closed)	Alpha 8–13 Hz, mainly occipital
N1	Theta 4–8 Hz, vertex sharp waves, slow eye movements
N2	Sleep spindles and K-complexes
N3 (slow-wave sleep)	≥ 20% high-amplitude delta (0.5–2 Hz)
REM	Low-voltage mixed frequency (like wake), rapid eye movements, low muscle tone, sawtooth waves

A night has 4–6 cycles of about 90 minutes. Automatic staging uses 30-s epochs, band powers, spindle detection and classifiers.

Epilepsy detection: features: spikes (20–70 ms), sharp waves (70–200 ms), 3 Hz spike-and-wave (absence seizures), rhythmic build-up in ictal periods. Methods: template matching, nonlinear energy

operators, line length, band power ratios, entropy, wavelets, machine-learning classifiers; seizure prediction from pre-ictal changes.

Spectral estimation methods

Method	Idea	Pros	Cons
Periodogram	$(1/N)\cdot\text{abs}(\text{DFT})^2$	Simple, FFT-based	High variance; leakage; resolution f_s/N
Averaged (Bartlett/Welch)	Average periodograms of segments (Welch: overlapping, windowed)	Lower variance	Poorer resolution
Blackman–Tukey	FT of windowed autocorrelation	Smooth	Resolution tied to lag window
Maximum entropy (Burg)	Extrapolate autocorrelation with maximum randomness	High resolution with short data	Model order choice; spurious peaks
AR model	All-pole model via Yule–Walker / Burg / covariance	Sharp peaks, short data	Order selection (AIC, FPE)
MA model	All-zero model	Good for deep valleys/notches	Poor for sharp peaks
ARMA model	Poles and zeros	Most general	Nonlinear estimation
Maximum likelihood (Capon / MLM)	Minimum-variance filter at each frequency	Better resolution than periodogram	More computation

AR model and Yule–Walker equations: for order p, $r(k) + \Sigma\, a_i\, r(k-i) = 0$ for $k = 1\ldots p$, and $\sigma^2 = r(0) + \Sigma\, a_i\, r(i)$. PSD: $P(f) = \sigma^2 / \left|1 + \Sigma\, a_k\, e^{-j2\pi fk/f_s}\right|^2$. Model order chosen by Akaike's information criterion (AIC) or final prediction error (FPE).

Why parametric for EEG: EEG is non-stationary, so short segments (1–2 s) are used; AR gives smooth, high-resolution spectra from short data and few parameters (also useful for compression and classification).

🧮 Solved examples

Example 1: Frequency resolution. EEG sampled at 256 Hz, 2-s segment (N = 512). Periodogram resolution = $f_s/N = 256/512 =$ **0.5 Hz**. A 1-s segment gives 1 Hz resolution.

Example 2: Band classification. A dominant 10 Hz rhythm in occipital leads with eyes closed is **alpha** (relaxed wakefulness). Dominant 2 Hz high-amplitude waves in sleep → **delta**, stage N3.

Example 3: AR(1) model. $r(0) = 1$, $r(1) = 0.8$. Yule–Walker (with $x(n) = a\,x(n-1) + e(n)$): $a = r(1)/r(0) =$ **0.8**; noise variance $\sigma^2 = r(0)(1 - a^2) = 1 \times (1 - 0.64) =$ **0.36**. PSD $= 0.36 / \left|1 - 0.8e^{-j\omega}\right|^2$; at $\omega = 0$: $0.36/0.04 = 9$; at $\omega = \pi$: $0.36/3.24 \approx 0.11$, a low-pass spectrum.

Example 4: Periodogram of a sinusoid. $x(n) = \cos(2\pi\cdot 10n/100)$, N = 100, $f_s = 100$ Hz. DFT bins are 1 Hz apart; energy appears at bins 10 and 90 (±10 Hz) with $|X| = N/2 = 50$, so $P = 50^2/100 =$ **25** at each.

Key points to remember

- Bands: delta < 4, theta 4–8, alpha 8–13, beta 13–30, gamma > 30 Hz.
- N2 = spindles + K-complexes; N3 = delta; REM = low-voltage mixed, eye movements.
- Epilepsy: spikes, sharp waves, 3 Hz spike-and-wave.
- Periodogram resolution f_s/N; Welch averages to cut variance.
- AR via Yule–Walker; MEM = Burg = AR; MA for notches; ARMA general; MLM = Capon.

⭐ Important questions

1. Explain EEG rhythms and their significance. Explain sleep EEG staging.
2. Explain EEG analysis for epilepsy detection.
3. Explain the periodogram and its limitations. How does Welch's method improve it?
4. Explain AR, MA and ARMA spectral estimation and the Yule–Walker equations.
5. Explain the maximum entropy and maximum likelihood methods of spectral estimation.

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UNIT 5 Evoked Potentials, Adaptive Filtering and Wavelets

✍️ 2-mark definitions

- **Evoked potential (EP):** the brain's electrical response to a specific stimulus (visual VEP, auditory AEP/BAEP, somatosensory SEP); very small (0.1–20 μV) and buried in background EEG.
- **Signal averaging (ensemble averaging):** adding N time-locked responses to a repeated stimulus; the EP adds coherently while random noise averages out.
- **SNR improvement by averaging:** for uncorrelated noise, amplitude SNR improves by √N (10 $\log_{10}$ N dB in power).
- **Adaptive filter:** a filter whose coefficients adjust automatically to minimize an error signal, useful when signal and noise statistics are unknown or changing.
- **LMS (least mean squares) algorithm:** $w(n+1) = w(n) + 2\mu\, e(n)\, x(n)$, a gradient-descent update for adaptive filter weights.
- **Step size (μ):** controls LMS speed and stability; $0 < \mu < 1/\lambda_max$ (often $0 < \mu < 1/(L \cdot P_x)$).
- **Adaptive noise canceller (ANC):** uses a reference input correlated with the noise (but not the signal) to estimate and subtract the noise from the primary input.
- **Wavelet transform:** represents a signal with scaled and shifted versions of a short wave (mother wavelet), giving time–frequency localization.
- **Mother wavelet:** the prototype wavelet ψ(t) (Haar, Daubechies db4, Mexican hat, Morlet).
- **Multiresolution analysis (MRA):** splitting a signal into approximation (low-pass) and detail (high-pass) coefficients level by level with filter banks.

💡 Core concepts

Signal averaging

- Model: $y_i(n) = s(n) + v_i(n)$, i = 1...N, where s is the repeatable EP and v_i is zero-mean noise uncorrelated across trials.
- Average: $\bar{y}(n) = s(n) + (1/N)\ \Sigma\ v_i(n)$; noise variance falls to σ^2/N.
- Requirements: stimulus-locked timing (trigger), stationary EP shape, noise independent of signal and between trials, zero-mean noise.
- Limits: latency jitter blurs the averaged EP; habituation changes the response; many trials take time. Remedies: Woody's method (align by cross-correlation), weighted averaging, moving averages.
- Uses: VEP (optic nerve), BAEP (hearing in infants, brainstem), SEP (spinal cord monitoring), P300 (cognitive, BCI).

Adaptive filtering and LMS

- FIR adaptive filter: output $y(n) = w^T x(n)$; error $e(n) = d(n) - y(n)$.
- Wiener solution $w^* = R^{-1}p$ (R = input autocorrelation, p = cross-correlation); LMS approaches it without knowing R and p.
- LMS steps: initialize w = 0; for each n compute y, e, update w. Simple, O(L) operations; convergence depends on μ and the eigenvalue spread. NLMS normalizes μ by input power; RLS converges faster

but costs more.

Adaptive noise cancelling (draw it): primary input d = s + n_o; reference input x = n_1 (correlated with n_o, not with s) → adaptive filter → y ≈ n_o; output e = d − y ≈ s. Minimizing E[e^2] minimizes noise power because s is uncorrelated with the reference.

Application	Primary	Reference
50 Hz removal in ECG	ECG + mains	Signal from mains outlet (sine and cosine)
Fetal ECG extraction	Abdominal lead (fetal + maternal ECG)	Chest leads (maternal ECG)
EOG removal from EEG	EEG + eye artifact	EOG electrodes
Heart sound noise removal	Lung sound mic	Heart sound reference
EMG in ECG, motion artifact	ECG	Accelerometer

Wavelet detection

- STFT uses a fixed window (fixed time–frequency resolution); wavelets use short windows for high frequencies and long ones for low frequencies, ideal for transients like QRS, spikes and EPs.
- CWT: $W(a, b) = (1/\sqrt{a}) \int x(t)\, \psi^*((t - b)/a)\, dt$ (scale a, shift b).
- DWT with filter banks (Mallat): at each level, low-pass + high-pass then downsample by 2 → approximation A_1 and detail D_1; repeat on A_1.
- At f_s = 360 Hz: D1 ≈ 90–180 Hz, D2 ≈ 45–90 Hz, D3 ≈ 22.5–45 Hz, D4 ≈ 11.25–22.5 Hz, D5 ≈ 5.6–11.25 Hz, A5 ≈ 0–5.6 Hz. QRS energy is mostly in D3–D5; baseline wander in deep approximations.
- **QRS detection with wavelets:** R peak = zero crossing between a modulus-maximum pair across scales; robust to noise and baseline.
- **Denoising:** decompose, threshold detail coefficients (soft/hard, universal threshold $\sigma\sqrt{(2 \ln N)}$), reconstruct.
- Other uses: spike detection in EEG, EP extraction from single trials, compression.

🧮 Solved examples

Example 1: Averaging gain. A VEP of 5 µV is buried in 50 µV RMS EEG noise (SNR = 0.1). With N = 100 trials, noise falls to 50/√100 = 5 µV, giving SNR = **1** (20 dB improvement). For SNR = 3: √N = 30 → **N = 900** trials.

Example 2: LMS update. Two-tap filter, w(0) = [0, 0], µ = 0.1, x(0) = [1, 0.5], d(0) = 2.

- y = 0, e = 2.
- w(1) = w(0) + 2 × 0.1 × 2 × [1, 0.5] = **[0.4, 0.2]**.
- Next sample x(1) = [0.5, 1], d(1) = 1: y = 0.4 × 0.5 + 0.2 × 1 = 0.4; e = 0.6; w(2) = [0.4, 0.2] + 0.12 × [0.5, 1] = **[0.46, 0.32]**.

Example 3: DWT frequency bands. f_s = 250 Hz (Nyquist 125 Hz). D1 = 62.5–125 Hz, D2 = 31.25–62.5 Hz, D3 = 15.6–31.25 Hz, D4 = 7.8–15.6 Hz, A4 = 0–7.8 Hz. Power line (50 Hz) lies in **D2**; the main QRS energy (≈ 8–25 Hz) in **D3–D4**.

Example 4: Haar one-level DWT. x = [4, 6, 10, 12]. Approximation (pairwise sums/√2) = [10/√2, 22/√2] ≈ [7.07, 15.56]; detail (differences/√2) = [−2/√2, −2/√2] ≈ [−1.41, −1.41].

Key points to remember

- Averaging: SNR × √N; needs time-locking, uncorrelated zero-mean noise, stable EP; Woody's method for jitter.
- LMS: w ← w + 2μex; NLMS, RLS.
- ANC: primary = signal + noise; reference = correlated noise only; output = error.
- Wavelets: good time resolution at high frequency; DWT filter bank halves the band each level.

⭐ Important questions

1. Explain evoked potentials and their estimation by signal averaging. Derive the SNR improvement.
2. Explain the LMS adaptive filter algorithm.
3. Explain adaptive noise cancellation with its applications in ECG and EEG.
4. Explain the wavelet transform and its use in QRS detection and denoising.
5. Compare STFT and wavelet transform for biomedical signals.

Last-hour checklist

☐ Signal classification; ECG/EEG/EMG/ERG amplitude and frequency table; acquisition difficulties; computers and image reconstruction.

☐ ECG waves and intervals; HR (75 bpm); QTc (0.40 s).

☐ Pan–Tompkins block diagram and equations; 30-sample window.

☐ Baseline wander and 50 Hz notch (y = [x(n) + x(n−2)]/2 at 200 Hz); ST analysis; arrhythmia table; Holter.

☐ TP (2:1), AZTEC (plateau/slope, CR 1.5 example), CORTES, Fan, Huffman (2.2 bits, H = 2.12), RLE, PRD.

☐ EEG bands; sleep stages; epilepsy features.

☐ Periodogram (0.5 Hz resolution); Welch; AR/MA/ARMA; Yule–Walker (a = 0.8, σ^2 = 0.36); MEM; MLM.

☐ Averaging (√N, N = 900); LMS ([0.4, 0.2]); ANC diagram and uses; wavelets and DWT bands.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Biomedical Digital Signal Processing	Willis J. Tompkins	PHI Learning	WorldCat

#	Title	Author(s)	Publisher	Where to get it legally
2	Biomedical Signal Processing: Principles and Techniques	D. C. Reddy	McGraw-Hill	WorldCat
3	Biomedical Instrumentation and Measurements	Leslie Cromwell, Fred J. Weibell, Erich A. Pfeiffer	PHI Learning	WorldCat
4	Biomedical Signal Processing, Vol. I (reference)	Arnon Cohen	CRC Press	WorldCat
5	Biomedical Signal Analysis: A Case-Study Approach (reference)	Rangaraj M. Rangayyan	Wiley	WorldCat
6	Medical Instrumentation: Application and Design (reference)	John G. Webster	Wiley	WorldCat

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